USER INTERFACE DESIGN EX-3

Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)

Command line Interface(CLI)

● A Text-based Interface (Command line interface) enables users to communicate with a computer using the command line, as opposed to using a graphical interface.

● Compact and Powerful – CLI is usually faster and more powerful than a GUI, particularly for expert users who know the correct commands.

● Automation & Scripting – The command-line interface is there to allow users to automate tasks with scripts, making the repetitive tasks easier.

● Lightweight and Resource Friendly – Compared to GUI application, CLI uses very little system resources which is why they are most suited for use on remote servers
example: Windows command prompt

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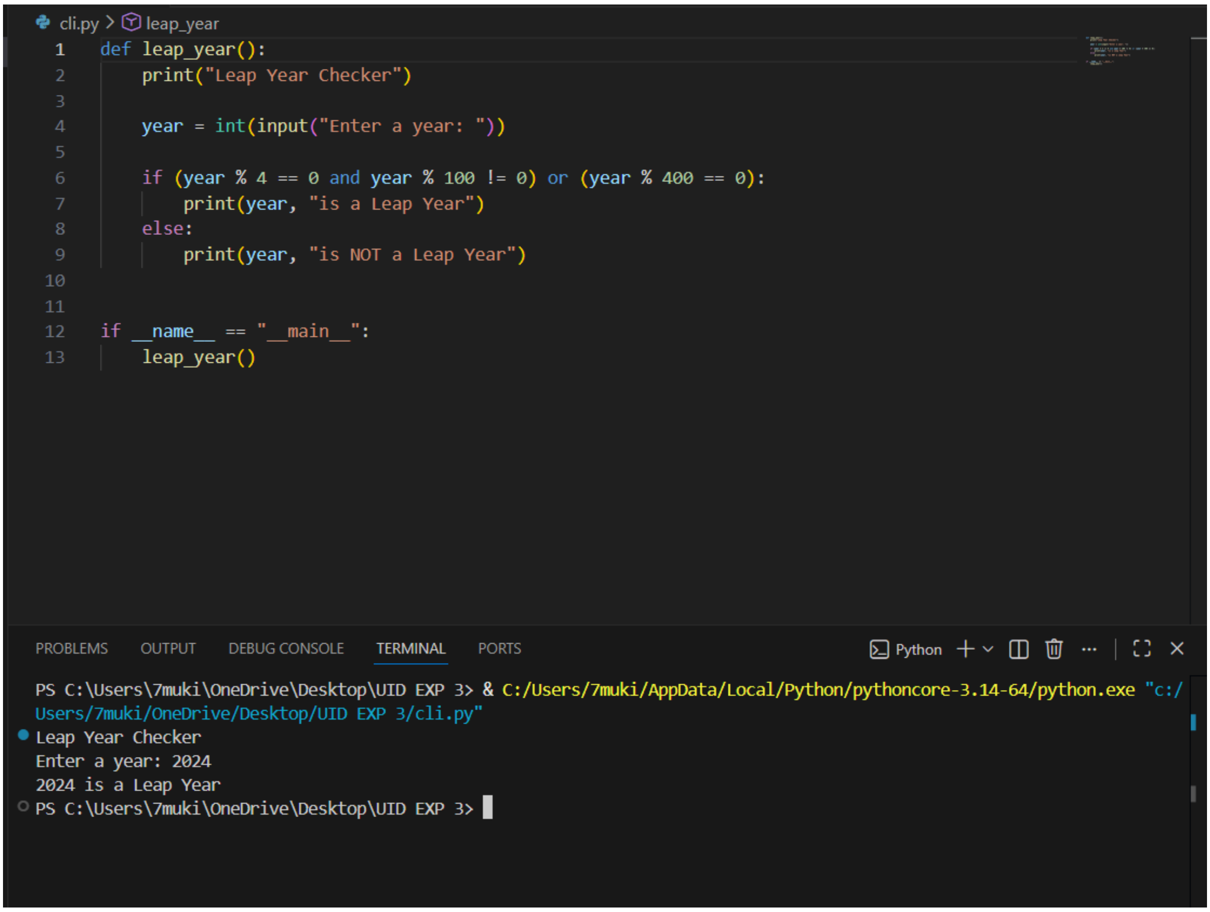
IMPLEMENTATION:



Graphical user interface(GUI)

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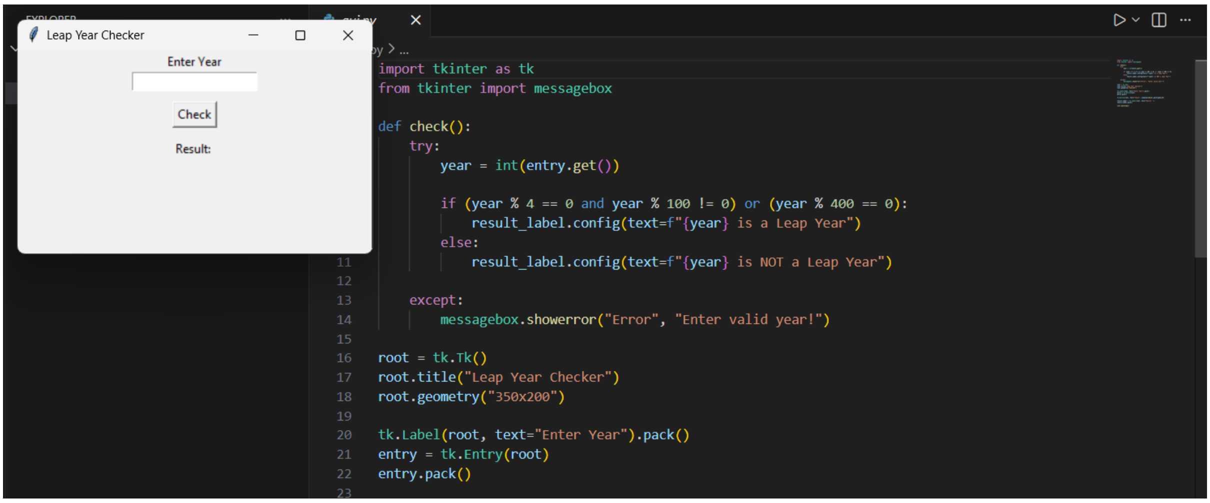
● A GUI uses visual elements including icons, buttons, and windows which makes it more user-friendly for users to work with the system.

● More Resource Intensive – GUI applications use more system resources (CPU, RAM, and GPU) in contrast to a Command Line Interface (CLI) that can degrade performance in low-end devices.

● Limited Automation: GUI does not have automation and scripting abilities like the CLI and is thus more tedious for repetitive jobs.
example: Windows, Android, iOS

IMPLEMENTATION:



Voice user interface(VUI)

• No Hands – VUI allows me to talk to a system and in some cases this is much more convenient than having to type.

• Natural Communication– VUI allows for more natural and fluid communication, as it can process users' speech and verbal commands, eliminating typing or clicking.

• Accuracy Challenges – Speech recognition can occasionally have difficulty with accents, background noise, and complex commands, sometimes resulting in misinterpretation.
example: Amazon, Alexa, Google Assistant

IMPLEMENTATION:

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```python
import speech_recognition as sr

def check_leap(year):
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
        print(year, "is a Leap Year")
    else:
        print(year, "is NOT a Leap Year")

def listen():
    recognizer = sr.Recognizer()

    with sr.Microphone() as source:
        print("Say a year (Example: 2024)...")
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        print("You said:", command)

        # Extract year from speech
        words = command.split()
        for word in words:
            if word.isdigit():
                year = int(word)
                check_leap(year)
                return

        print("No valid year detected.")

    except sr.UnknownValueError:
        print("Could not understand audio.")
    except sr.RequestError:
        print("Check your internet connection.")

if __name__ == "__main__":
    listen()
```

```
PS C:\Users\7muki\OneDrive\Documents\Codewave-software-website-main> & C:/Users/7muki/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/7muki/OneDrive/Documents/Codewave-software-website-main/backend/models/cli.py
Leap Year Checker
Enter a year: 2007
2007 is NOT a Leap Year
```